

Model Sense

ALM Model Lab — User Guide

A browser-based teaching lab for interest-rate simulation, behavioral instrument modeling, and asset-liability analytics. This edition pairs the operating manual with the concepts, functional forms, and parameter defaults of every model.

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Orientation

What the Lab is, how the four sections fit together, the order to work through them, and how to read the technical material.

What the Lab does

The ALM Model Lab takes a market snapshot, calibrates two interest-rate models to it, simulates forward rate paths, and prices a small book of bank instruments and behavioral deposits against those paths. Everything runs in your browser; nothing is uploaded.

It is a teaching tool. The numbers are illustrative and the defaults are pedagogical, not a production risk system. The point is to make the mechanics of rate simulation, prepayment, deposit behavior, replicating portfolios, and funds transfer pricing visible and manipulable.

The four sections, in working order

The top navigation has four sections. Work left to right the first time through.

- **Market Data.** Import a snapshot, review the rate history, bootstrap the SOFR curve, and inspect the volatility surface (SABR). Calibration of the rate models happens here.
- **Rate Models.** Run the Hull-White and BGM simulations and compare them. These produce the path sets every downstream analytic consumes.
- **Instruments.** Parameterize the fixed loan, floating loan, mortgage, the two non-maturity deposit books, and build the deposit replicating portfolio.
- **ALM Analytics.** Repricing gap, liquidity gap, funds transfer pricing (loans and NMD), effective duration, and the stochastic earnings gap. This is where the book-level results live.

The typical workflow

A full pass is: import a snapshot on the Import tab, open the SABR tab and click Calibrate models, then run the Hull-White and BGM simulations on their tabs. With both simulations in hand, every Instruments and Analytics tab is live. The FTP and Replicating Portfolio tabs need both simulations because they optimize and option-price over the simulated paths.

The header shows the loaded calibration date and path count. The info button (top right) opens a popover with the calibrated model parameters and known limitations.

How to read the technical material

Each section below pairs the operating instructions with the functional form and the parameter defaults of the models that drive it. Read the Notation and conventions section first: it fixes the units, the sign convention, the rate-path bridge, and the two shock conventions that the rest of the guide assumes throughout.

Formulas are written exactly as the engine computes them. Where the Lab simplifies a canonical method for tractability or teaching, the simplification is stated rather than hidden. Defaults are the values shipped on the tabs; almost all are editable.

Notation and conventions

Units, signs, the rate-path bridge, the Monte Carlo defaults, and the EVE-versus-NII shock conventions that every model below assumes.

Units and sign conventions

The Lab mixes annualized rates with monthly behavioral intensities by design, so units are load-bearing. Read a formula by first checking what each quantity is denominated in.

- **Curve rates.** Decimal per annum ($0.04 = 4\%$). Zero rates are continuously compounded; discount factors are $P(t) = \exp(-z(t) \cdot t)$.
- **Deposit and decay intensities.** Closure and rate-driven decay are in percent per month and are divided by 100 before they compose; deposit rates and spreads are in percent per annum.
- **Time grid.** Monthly. A cashflow at month m sits at $t = m/12$ years. The fixed-leg day-count is ACT/360, applied as 365/360 per full year.
- **Balances.** Behavioral runoff fans are normalized to a start of 100; cashflow and PV tables are in dollars.
- **Sign.** Assets carry a positive sign and liabilities negative in the gap and SEG views. FTP margins are constructed to be positive franchise value on both sides: asset NIM is yield minus all-in FTP, liability credit is all-in FTP minus deposit cost.

The rate-path bridge

Every instrument is priced against a RatePath: an object exposing a short rate at each monthly step and a term forward at any tenor. A deterministic path reads the central bootstrapped curve; a stochastic path wraps a single Hull-White or BGM realization. The same generator code runs in both, so the deterministic line and the Monte Carlo mean are produced identically.

The one subtlety a practitioner must hold: term forwards beyond one year are read as par-swap rates, not as simple-compounded money-market forwards, because simple compounding overstates multi-year rates by hundreds of basis points. The split happens at one year.

$$f(t1, t2) = (P(t1)/P(t2) - 1) / (t2 - t1)$$

simple-compounded forward, money-market convention; used only for tenors up to 1Y

$$S(t1, t2) = (P(t1) - P(t1+n)) / \sum_k (365/360) \cdot P(t1+k), \quad n = \text{round}(t2 - t1)$$

forward par-swap rate, annual ACT/360 fixed leg; used for tenors beyond 1Y

Monte Carlo defaults and the martingale correction

Both simulators run 500 paths as 250 antithetic pairs over a 30-year monthly horizon from a fixed seed, so a given snapshot reproduces bit-for-bit. Antithetic pairing draws one standard normal per pair and applies it with both signs, halving the variance of any quantity that is roughly linear in the shock.

After simulation, a multiplicative correction rescales the per-step discount factors so the simulated mean discount factor equals the market discount factor at every horizon. This removes the small numerical drift of the discretized scheme; it is distinct from the analytic convexity term inside the bond formula. The raw, pre-correction error is reported as a diagnostic.

$$\text{correction}_k = P^M(0, t_k) / E_p[\tilde{D}_p(t_k)]$$

P^M is the market discount factor; E_p the path-mean of the uncorrected simulated discount factor at step k

Two shock conventions: EVE versus NII

A parallel rate shock means two different things depending on what you are measuring, and the Lab keeps them separate on purpose.

For an economic-value (EVE) or effective-duration view, the shock applies from step 0: the entire rate environment moves instantly, including the first coupon. For an earnings (SEG / NII) view, the first month's coupon is locked before the shock event, so step 0 is left unshocked. The discount-side shock is a parallel shift of the zero curve; the cashflow-side shock re-simulates the behavioral response, so prepayment and deposit beta react to the new path rather than merely being re-discounted.

Market Data

Import, rate history, curve bootstrap, the term liquidity premium, and the SABR volatility surface.

Import

Loads a market snapshot: the SOFR curve quotes, cap and swaption volatilities, FHLB advance curve, and the term liquidity premium nodes. A default snapshot loads automatically so you can explore without a file. You can drop your own workbook to override it.

Cap and swaption volatilities are normal (Bachelier, basis-point) vols, not lognormal. Cap strikes are absolute rates and may be negative, reflecting the negative-rate-era convention; the at-the-money cap is carried as its own column

rather than inferred from a zero strike.

Rate History and the deposit-tractor pillar

The historical SOFR term series drives the replicating-portfolio moving averages. The steady-state yield of a k-month pillar is the trailing k-month average of the k-month tenor, so the history is what gives the deposit tractor its carry lag.

Two details matter when reconciling against the replicating portfolio. The pillar moving average is inclusive of the as-of month, which differs from the strictly-lagged window the Non-IB deposit model uses for its regime moving average; do not conflate the two. And the series is a stylized FOMC-cycle reconstruction, not literal SOFR, since SOFR did not exist before 2018; the pre-2018 segment is an OIS/EFFR proxy splice.

$$\text{trailingMA}(\text{values}, i, k) = (1/k) \cdot \sum_{j=i-k+1 \dots i} \text{values}[j]$$

inclusive trailing k-month average at index i; NaN before the first complete window

$$\text{pillarYield}(\text{asOfIdx}, k) = \text{trailingMA}(\text{sofrTermSeries}(k), \text{asOfIdx}, k)$$

steady-state yield of the k-month tractor pillar, decimal per annum

Curve

Bootstraps the SOFR zero curve from the snapshot quotes. Short-end nodes at or below one year invert directly from a simple-compounded discount factor; longer nodes are solved one at a time so each maturity reprices to par given the shorter nodes already fixed. The solve is an exact root-find per node, not a least-squares fit, and interpolation is linear in the zero rate, which makes a parallel shock exact at every node.

Every downstream analytic (FTP, EVE, cap pricing, the simulation drivers) reads this one curve, so there is a single object of truth. The float leg uses the textbook par identity rather than projecting SOFR fixings, the standard single-curve OIS simplification where the discount and forward curves coincide.

$$df = 1 / (1 + r \cdot t), \quad z = -\ln(df) / t$$

short-end inversion, tenor t up to 1Y; r is the quoted simple rate, z the continuously-compounded zero

$$1 - \exp(-z_N \cdot t_N) = \sum_k r \cdot (365/360) \cdot \exp(-z(c_k) \cdot c_k)$$

par-swap node equation for tenors beyond 1Y; coupon dates $c_k = 1..n$ with $n = \text{round}(t_N)$; solve z_N by Brent root-find

$$\text{shockCurve}: z(t) \rightarrow z(t) + \text{shockBp}/1e4$$

the EVE discount-side shock; a parallel shift of every node, so discount factors scale by $\exp(-s \cdot t)$

Term liquidity premium and the all-in funding curve

The term liquidity premium (TLP) is the spread of FHLB advance rates over matching-tenor SOFR: the term funding cost a desk actually pays. It is pinned to zero overnight, since the bank funds overnight at policy rates rather than through advances, and rises with tenor. It composes additively with the SOFR zero curve to form the all-in funding curve that FTP charges and credits against, which cleanly separates the pure-rate leg from the liquidity leg.

The shipped node set is the 9/30/2025 FHLB-minus-SOFR spread, interpolated linearly and held flat outside the bracket. The empirical short-end dip (3M below 1M) is preserved rather than smoothed.

$$z_{\text{all-in}}(t) = z_{\text{sofr}}(t) + \text{tlp}(t)$$

all-in funding curve; TLP applied to discounting as $\exp(-\text{tlp}(t) \cdot t)$

- **Node set.** Overnight 0, 1M 21 bp, 3M 19 bp, 6M 20 bp, 1Y 31 bp, 2Y 43 bp, 3Y 50 bp, 5Y 56 bp, 10Y 91 bp, 30Y 153 bp. Linear in spread, clamped flat at the ends.

SABR volatility surface

The SABR tab shows the normal-vol smile and triggers the two model calibrations. The smile is the Hagan 2002 normal-vol expansion in its beta-equals-zero (Bachelier) limit, evaluated from a parameter set chosen by a strike-coverage heuristic rather than fit by least squares, so treat the displayed surface as a smile interpolant, not a market-fitted SABR. The genuine least-squares fits are Hull-White to the at-the-money cap column and BGM to the swaption surface, both run when you click Calibrate models. The rate-model tabs stay disabled until that calibration has run.

$$\sigma_N(ATM) = \alpha \cdot (1 + ((2 - 3\rho^2) \cdot \nu^2 / 24) \cdot T)$$

at-the-money normal vol; α is the level (vol of rate), ν the vol-of-vol, ρ the skew, T the expiry

$$z = (\nu/\alpha) \cdot (F - K), \quad x = \ln(\sqrt{(1 - 2\rho z + z^2)} + z - \rho) / (1 - \rho)$$

away-from-ATM; F is the forward, K the strike

$$\sigma_N = \alpha \cdot (z/x) \cdot (1 + ((2 - 3\rho^2) \cdot \nu^2 / 24) \cdot T)$$

the ATM branch is the $z \rightarrow 0$ limit where $z/x \rightarrow 1$

Rate Models

Hull-White and BGM simulation, their calibration and dynamics, and the engine comparison.

Hull-White (1-factor)

An exact-transition short-rate simulation. The short rate is the bootstrapped instantaneous forward plus a mean-reverting Gaussian factor, so the model is curve-consistent at every horizon by construction. The factor follows an Ornstein-Uhlenbeck process that is sampled by its exact one-step transition, so there is no time-discretization bias in the factor itself; the only discretization is the trapezoidal accrual used for discounting, which the martingale correction then removes.

Hull-White projects any tenor analytically from its single latent state through the affine bond formula, which is why the option-cost and replicating-portfolio machinery can reconstruct term rates cleanly without Monte Carlo noise. Calibration is a Levenberg-Marquardt least-squares fit of the mean reversion a and volatility σ to the at-the-money cap column, with the residual measured in basis points; the smile is left to the SABR display. The rate can go negative, which is correct for a Gaussian model.

$$r(t) = f^M(0, t) + X(t)$$

short rate = market instantaneous forward plus the mean-reverting factor; $X(0) = 0$

$$X_{k+1} = X_k \cdot \exp(-a \cdot dt) + \sigma \cdot \sqrt{(1 - \exp(-2a \cdot dt)) / (2a)} \cdot Z$$

exact OU transition; Z standard normal, applied with both signs across an antithetic pair

$$B(a, \tau) = (1 - \exp(-a \cdot \tau)) / a$$

the factor loading; the building block of every Hull-White bond and vol formula

$$P(t, T | X) = (P^M(0, T) / P^M(0, t)) \cdot \exp(-B(\tau) \cdot X(t) - V), \quad V = (\sigma^2 / 4a) \cdot (1 - \exp(-2a \cdot t)) \cdot B(\tau)^2$$

shifted-Gaussian bond reconstruction, $\tau = T - t$; arbitrage-free and curve-consistent at $t = 0$

- **a (mean reversion)**. Speed of pull-back to the forward curve, per year. Calibrated; bounded [1e-4, 0.5].
- **sigma (volatility)**. Absolute (normal) volatility of the short rate, decimal per square-root year. Calibrated; bounded [1e-6, 0.05].
- **Simulation**. 500 paths (250 antithetic pairs), 30-year monthly horizon, fixed seed, multiplicative martingale

correction on.

BGM / LMM (2-factor)

A LIBOR market model that evolves the forward rates directly under the spot-LIBOR measure. The instantaneous volatility of each forward is the Rebonato four-parameter parametric form, which depends only on time-to-maturity and so is time-homogeneous; two orthogonal loadings rotated by an angle β give the two factors, and the forward-forward correlation is the cosine of the rotation difference. This compact structure is what lets a two-factor model fit the whole swaption surface.

Simulation is in Glasserman-Zhao log coordinates with optional displaced-diffusion and CEV dynamics, advanced by a predictor-corrector step that averages the drift over the start and end of the period using the same shocks. The displacement and CEV exponent tame the long-horizon upper tail, which is otherwise fuller than Hull-White's. Calibration is a Levenberg-Marquardt fit of the volatility parameters and rotation angle to the at-the-money swaption grid, accepted near or below ten basis points RMSE; because the objective is multimodal, the fit runs from two starting points and keeps the better. The displacement and CEV exponent are exogenous structural inputs, not calibrated.

$$\sigma_i(t) = \text{volScalar} \cdot [(a + b \cdot (\tau_i - t)) \cdot \exp(-c \cdot (\tau_i - t)) + d]$$

Rebonato instantaneous vol of forward i ; $\tau_i - t$ is time-to-maturity

$$\rho_{ij} = \cos(\beta \cdot (\tau_i - \tau_j))$$

time-homogeneous forward-forward correlation from the rotation angle β

$$\sigma_N(ATM) \approx \sqrt{(\sum_i \sum_j w_i \cdot w_j \cdot \hat{F}_i^{\beta_{cev}} \cdot \hat{F}_j^{\beta_{cev}} \cdot \rho_{ij} \cdot I_{ij} / T_{\alpha})}$$

frozen-weight swaption normal vol; $\hat{F} = F + \delta$ (displacement), w the annuity weights, I_{ij} the closed-form vol-product integral

$$d \log \hat{Y}_k = (\mu_k - \frac{1}{2} \sigma_{loc}^2) \cdot dt + \sigma_{loc} \cdot d\tilde{W}, \quad \hat{Y}_k = F_k + \delta, \quad \sigma_{loc} = \hat{Y}^{\beta_{cev} - 1} \cdot \sigma$$

log-coordinate dynamics; $\beta_{cev} = 1$ collapses to displaced-lognormal

- **a, b, c, d.** Rebonato vol shape: $a + b \cdot \text{tenor}$ humped by $\exp(-c \cdot \text{tenor})$ plus a floor d . Calibrated jointly.
- **beta (rotation).** Rotation angle setting the two-factor correlation, in radians; bounded $[-1, 1]$. Distinct from the CEV exponent.
- **displacement and CEV exponent.** Structural tail controls (delta shift, β_{cev} elasticity), chosen exogenously; the vol parameters are re-fit after they are set.

Compare

Side-by-side fan charts of the two engines at a chosen tenor, so you can see where they agree and where the BGM tail widens. The FTP and Replicating Portfolio tabs let you choose which engine drives the result; this tab is where you build intuition for that choice. Both engines pool onto the same path-and-month grid downstream, so their replicating-portfolio blends are directly comparable.

Instruments

Loans, the prepayable mortgage, the two NMD books, the deposit tractor, and their functional forms.

Fixed and floating loans

The fixed loan amortizes at a locked coupon; its cashflows are rate-path independent by construction, which is the definition of fixed. The floating loan reads the index in arrears, resets at its cadence, and re-amortizes the remaining balance over the remaining term at the new coupon, which is the practitioner-correct treatment: a rate move changes the payment, not the schedule length. Both are the clean reference cases for FTP: a fixed coupon

match-funds at a fixed all-in rate, a floater funds at the reset-tenor SOFR plus TLP.

$$PMT = B_0 \cdot i \cdot (1+i)^N / ((1+i)^N - 1), \quad i = coupon / 12$$

level-pay monthly payment on notional B_0 over N months; straight-line B_0/N in the zero-rate limit

$$couponRate_m = r(t_{m-1}) + margin \quad \text{on reset months } m \in \{1, 1+R, 1+2R, \dots\}$$

floating coupon, index read in arrears at the prior month; R is the reset frequency in months

Mortgage and the prepayment model

A level-pay mortgage with a four-factor prepayment model. The note rate is the ten-year forward plus a fixed origination spread; prepayment is rate-path-dependent through the conditional prepayment rate (CPR), which is what gives the mortgage its embedded short option and makes its stochastic FTP differ from the single-path view. The CPR is the product of four factors: a refinancing-incentive logistic in the rate difference, a seasoning ramp, a seasonal sinusoid, and a burnout that decays with cumulative in-the-moneyness. The annualized CPR converts to a monthly single-month mortality (SMM) by the exact geometric relation, and SMM is applied to the balance net of scheduled principal.

The prepayment factors are balance-independent, which licenses a single-pass schedule. A Richard-Roll arctan refinancing variant is also implemented for comparison; it takes decimal rates and differs in the seasonality and burnout scaling, so the two are not interchangeable without rescaling.

$$rateDiff = wac - mortgageRate \cdot 100$$

refinancing incentive in percentage points; wac is the pool note rate in points, $mortgageRate$ the current rate in decimal

$$refi = (minCpr + (maxCpr - minCpr) / (1 + \exp(-steepness \cdot (rateDiff - inflection)))) / 100$$

the refinancing-incentive S-curve, output as a decimal CPR

$$seasoning = \min(1, age/ramp), \quad seasonality = 1 + (amp/100) \cdot \sin(\pi \cdot (monthIndex - 4)/6)$$

the age ramp and the annual seasonal factor

$$burnout = \exp(-(burnoutDecay/100) \cdot \Sigma \max(0, rateDiff))$$

decays prepayment with cumulative positive in-the-moneyness

$$CPR = \text{clip}(refi \cdot seasoning \cdot seasonality \cdot burnout, 0, 1), \quad SMM = 1 - (1 - CPR)^{(1/12)}$$

the four-factor product, clamped, then converted to a monthly mortality

- **wac / steepness / inflection.** Pool note rate (6.5 points), logistic steepness (2.5), and inflection (0.5 points). The S-curve midpoint sits where the refinancing incentive is half its range.
- **minCpr / maxCpr.** Floor and ceiling annual CPR, 2% and 65%, before the seasoning, seasonality, and burnout overlays.
- **seasoningRamp / seasonalityAmp / burnoutDecay.** Months to full speed (30), seasonal amplitude (20%), and burnout decay rate (0.1).

Non-IB NMD: the decay model

The non-interest-bearing book pays no coupon and runs off on a behavioral decay model. Two limbs compose multiplicatively each month: an age-ramped baseline closure that captures relationship attrition, and a one-sided rate-driven incentive that captures depositors leaving when the market out-pays the parked balance. The incentive is a logistic in the spot rate $LEVEL$ — the opportunity cost a 0%-paying deposit forgoes — multiplied by a regime amplifier that accelerates runoff when the rate has jumped above its trailing moving-average regime (the surge of low-yield balances now repricing out), an optional Schwartz-Torous burnout, and the balance-sensitivity multiplier $g(b)$. The amplifier is the SSRN excursion form $1 + \alpha \cdot \max(0, r - MA)$: dormant at or below the prevailing regime, it captures the path dependency that a flat rate-level model misses.

Rate sensitivity is graded by account size through $g(b)$, a logistic in log-balance: every dollar is rate-sensitive, but large accounts leave more opportunity cost on the table at a low deposit rate and so run off faster. The book is partitioned into editable account-size buckets, each running the same closure and incentive at its own $g(\bar{b}_i)$, and the portfolio decay is the balance-weighted aggregate. The decay surface is the outer product of the balance logistic $g(b)$ and the rate-level logistic, the two factors visible on the tab's 3D surface. The model is explicitly illustrative; the canonical Model Sense deposit framework is separate.

$$MA_P(t) = (1/P) \cdot \sum_{j=1..P} r_1M(t-j), \quad excursion(t) = \max(0, r_1M(t) - MA_P(t))$$

trailing regime moving average and its one-sided excursion, percent per annum; the MA uses the strictly prior P months

$$base_incentive(t) = \maxGrowth + (\maxDecay - \maxGrowth) / (1 + \exp(-K \cdot (r_1M(t) - midpoint)))$$

opportunity-cost logistic on the spot rate LEVEL, percent per month; maxGrowth = 0 makes it run from 0 to maxDecay

$$g(b) = g_min + (g_max - g_min) / (1 + \exp(-\kappa_b \cdot (\ln b - \ln b^*)))$$

balance-sensitivity multiplier, logistic in log-balance; b^* the inflection balance

$$rateIncentive_i(t) = \max(0, base_incentive(t) \cdot regimeAmp(t) \cdot B(t) \cdot g(\bar{b}_i))$$

regimeAmp(t) = $1 + \alpha \cdot excursion(t)$ the SSRN surge amplifier; $B(t) = \exp(-\lambda_b \cdot \sum_{s<t} \max(0, base_incentive(s)))$ the burnout

$$closure(t) = closure_steady + (closure_initial - closure_steady) \cdot \exp(-(t-1)/\tau)$$

age-ramped baseline closure, percent per month

$$decay_i(t) = 1 - (1 - closure(t)/100) \cdot (1 - rateIncentive_i(t)/100)$$

combined monthly decay fraction per bucket; portfolio decay is the balance-weighted aggregate

- **closure_steady / closure_initial / tau.** Long-run closure 0.75%/mo, initial 2.0%/mo, settling with a 24-month time constant. Initial equals steady disables the ramp.
- **maxRateDecay / maxRateGrowth / K / midpoint.** High-rate plateau 1.98%/mo (re-solved so the seasoned-stock WAL lands near 5 years on the 3/31 curve), growth 0 (one-sided), steepness 1.5, midpoint 1.0% rate level (well below the 3/31 operating level of ~3.7%, so the incentive is near its plateau and the level-based logistic produces a modest convexity wedge: effective duration < static discounting duration < WAL).
- **regime alpha / burnout lambda.** Regime amplification $\alpha = 2.0$ (SSRN-calibrated): the surge amplifier $1 + \alpha \cdot \max(0, r - MA_{24})$ is dormant at or below the trailing regime and amplifies in a rate spike. Burnout off by default (lambda 0).
- **g(b): g_min / g_max / b* / kappa_b.** Floor 0.15, ceiling 1.0, inflection \$1.1M, steepness 1.3. Shared with the Replicating Portfolio tab, where the account-size buckets read the same shape.

IB NMD: deposit beta and the rate path

The interest-bearing book layers a dynamic deposit rate on top of the same closure decay model. The deposit rate tracks the one-month market index as a level ratio through a logistic deposit-beta S-curve, with an optional Nerlove partial adjustment that smooths the path toward target. The closure overlay then runs against the spread of market rate minus deposit rate rather than the rate-surprise spread, with the logistic midpoint shifted to re-center on the typical interest-bearing regime. The book skews toward large, rate-aware commercial balances, so it sits high on $g(b)$ and runs off faster than the retail Non-IB book.

The default beta parameters are the in-sample MMDA estimates from Chen (2026), Dynamic Deposit Betas (SSRN 6269838), not free knobs. The Nerlove factor defaults to full pass-through; the paper estimates it near 0.47, which smooths the trajectory when set in that range. A single modeled deposit-rate path is shared across all vintages, so the FTP and margin views reuse one consistent rate.

$$\beta(r) = \beta_min + (\beta_max - \beta_min) / (1 + \exp(-k \cdot (r - m)))$$

deposit-beta S-curve; r the one-month rate in percent, m the inflection level

$$D_target(t) = \beta(r(t)) \cdot r(t)$$

level-ratio target deposit rate, percent per annum

$$D(t) = \max(0, D(t-1) + \lambda \cdot (D_target(t) - D(t-1))), \quad D(0) = D_target(0)$$

Nerlove partial adjustment; $\lambda = 1$ snaps to target every period

$$spread(t) = r(t) - D(t)$$

the switching incentive feeding the IB closure overlay; Non-IB uses the spot rate level directly, IB uses this spread

- **beta_min / beta_max.** Lower and upper beta asymptotes, 0.433 and 0.800 (Chen 2026, in-sample MMDA).
- **k / m.** Logistic steepness 0.566 and inflection 3.919%.
- **lambda (Nerlove).** Partial-adjustment speed in (0, 1]; default 1.0 (full pass-through), paper estimate near 0.47.
- **closure overlay.** Same form as Non-IB with the midpoint shifted to 2.0 (re-centered on the $r - D$ regime) and a higher rate-decay plateau (9.5%/mo), reflecting the commercial book's faster flight. The IB closure overlay carries no rate-level multiplier in its base form: the regime amplifier is present but off by default ($\alpha = 0$, retained for stress exploration), and the old cash-sorting salience has been dropped entirely — the beta coupon already prices in the rate level through $D = \beta(r) \cdot r$. The base rate response is the $r - D$ spread alone. The decay-incentive steepness is flattened (logisticK = 0.3): at the steeper $K = 1.0$ the runoff over-responded to the ± 100 bp shock, pushing the effective duration slightly negative on the stochastic legs (a deposit liability cannot have negative duration). Flattening it holds the rate-locked $(1 - \beta) \approx 20\%$ slice near a 2-year duration, landing a stable, positive effective duration of ~ 0.25 to ~ 0.40 years across the deterministic and stochastic scenarios, consistent with the $(1 - \beta) \cdot 2y \approx 0.40$ economic anchor; the deterministic-forward WAL eases to ~ 2.4 years. Under the ± 100 bp shock the coupon reprices at $\beta(r \pm \Delta) \cdot (r \pm \Delta)$, and the effective deposit beta including the slope term $\beta_{eff} = \beta + r \cdot \beta'(r) \approx 0.79$ at the 3/31 operating rate.

Replicating Portfolio (the deposit tractor)

Represents the stable deposit balance as a ladder of moving-average pillars. A k -month pillar is k equal rolling bullets in the k -month tenor: a linear-amortizing runoff with $WAL = (k+1)/2$ months earning the trailing k -month average of the k -month rate. Two yields are tracked and never mixed: the steady-state carry (the trailing moving average) and the new-money yield (today's curve par). The overnight bucket is the spot rate with a half-month WAL by convention, not a one-month bullet.

The optimal blend is solved on the simulated paths as a mean-variance efficient frontier, where risk is the volatility of monthly yield changes. For the non-interest-bearing book it is a yield-space frontier (maximize yield per unit yield-volatility under a WAL cap, with the Sharpe ratio measured against the forecast-mean overnight yield). For the interest-bearing book it is a margin-space frontier that nets the modeled client rate, plus a non-negative least-squares regression of the deposit rate on the pillar yields constrained to sum to one. A Hull-White or BGM toggle selects which engine's paths drive the frontier.

This tab is an interest-rate view only: it builds the portfolio and prices its credit on term yields and repricing, with no liquidity overlay. The contingent-liquidity stress and HQLA buffer charge live on the FTP - NMD tab. The Non-IB section adds a stochastic credit fan: because the client rate is zero, the credit dispersion across paths is the structural interest-rate risk, and a shorter blend reprices faster and widens the fan.

$$WAL_pillar(k) = (k + 1) / 2 \quad \text{for } k > 1, \quad 0.5 \quad \text{for } k \leq 1$$

weighted-average life of a k -month moving-average pillar, in months

$$\Sigma_{ij} = 12 \cdot \text{Cov}(\Delta L_i, \Delta L_j)$$

annualized covariance of monthly pillar-yield changes; the frontier risk matrix, ridge-regularized in the solver

$$\min w' \Sigma w \quad \text{s.t.} \quad \Sigma w_i = 1, \quad \Sigma w_i \cdot WAL_pillar(k_i) \leq cap, \quad w \geq 0$$

the min-variance blend under the WAL cap; max-Sharpe is the cloud point maximizing $(ret - rf)/vol$

$$IB \text{ margin: } ret = \Sigma w_i \mu_{Y,i} - \mu_D, \quad Var(m) = w' \Sigma_Y Y w - 2 w \cdot \Sigma_Y D + Var(D)$$

the cross term $-2 w \cdot \Sigma_Y D$ is hedge value: the min-margin-vol blend tracks the client rate

- **Corner blends.** Min-vol, max-Sharpe, max-yield (or max-margin), and Stock-WAL cap. Pick one to drive the stock-view runoff and flow-view new-money ladder.
- **Stock vs flow.** Stock view is the portfolio's expected monthly amortization (drives repricing gap and EVE). Flow view is what each month's reinvestment buys as new bullets; shorter pillars draw proportionally more new money.
- **WAL cap.** The frontier binds at the cohort-derived stock WAL (the behavioral anchor). The liquidity WAL cap is shown as a separate reference overlay, never used to fund the liquidity gap off the portfolio.

Seasoned stock versus the fresh cohort

The instrument tabs model a single fresh cohort, but a real deposit book is a stack of vintages of different ages. The Lab superposes the cohort survival kernel across a steady-state age distribution to recover the seasoned stock, and reports both the stock WAL and the cohort WAL. The stock is always longer-lived than a fresh cohort, and this is a structural consequence rather than an assumption: closure declines with age, so survivors are stickier, and the survivor-weighted stock runs off more slowly than the cohort it was drawn from. On a flat rate path the stock runoff reduces to the renewal-theory identity.

$$B[m] = \sum_{size} \sum_{age} (sizeShare \cdot ageShare) \cdot surv_{\{size,age\}}[m]$$

seasoned-stock balance as a size-by-age superposition of cohort survivals, normalized to 1

$$R(t) = \sum_{\{j \geq t\}} S(j) / \sum_{\{j \geq 0\}} S(j)$$

the flat-path renewal identity: remaining stock fraction at month t under constant origination

$$WAL = \sum_{\{m\}} runoff_m \cdot (m/12) + B[N] \cdot (N/12)$$

stock WAL in years, including the residual tail at the horizon to avoid a short bias

ALM Analytics

Gap views, the two FTP tabs, effective duration and economic value, and the stochastic earnings gap.

Repricing Gap and Liquidity Gap

Repricing gap buckets balances by when they reprice; liquidity gap buckets them by when principal actually runs off. Each exports to Excel with a summary plus one sheet per instrument.

FTP - Loans

Marginal funds transfer pricing on new volume for the fixed loan, floating loan, and mortgage. Each instrument is priced as new origination match-funded against the all-in curve: par-matched against SOFR alone (interest-rate FTP), against SOFR plus TLP (all-in FTP), and the residual is the liquidity-premium FTP. The par-match is linear in the transfer rate, so it is a direct algebraic solve with no root-finding and no optionality on a single path. The FTP margin is the locked NIM, asset yield minus all-in FTP.

The mortgage also gets a stochastic, option-adjusted view. Choose Hull-White or BGM, then run it. The Lab regenerates the mortgage cashflows along every simulated path and path-averages the par-coupon FTP; averaging the linear par-rate functional across paths is exact and preserves the covariance between discounting and cashflow timing that root-finding per path would drop. The vol value is the option premium a single forward path misses, measured as a double difference per leg. Z-spread versus OAS, both over SOFR, cross-checks the interest-rate-leg vol value.

$$r = (N - \sum_i D(t_i) \cdot P_i) / (\sum_i D(t_i) \cdot B_{\{i-1\}} \cdot \Delta t), \quad \Delta t = 1/12$$

par-matched transfer rate; D the discount function, P_i principal, B_{\{i-1\}} start-of-month balance

$$D_{IR}(t) = P_{sofr}(t), \quad D_{all-in}(t) = P_{sofr}(t) \cdot \exp(-TLP(t) \cdot t), \quad lpFtp = allInFtp - irFtp$$

the two discount bases; the liquidity-premium FTP is the additive residual

$$volValue_{leg} = (stoch_{on} - stoch_{off}) - (det_{on} - det_{off})$$

double difference per leg (IR, LP, all-in); 'off' forces CPR to zero, isolating the convexity a single path cannot represent

$$optionCost = Z - OAS$$

Z-spread on the deterministic prepay-on path, OAS on the per-path mean, both over SOFR; the difference is the IR-leg option cost

FTP - NMD

Portfolio-level FTP on the standing deposit book under a constant-balance-sheet assumption, not marginal new-volume pricing. It is a snapshot of the stock FTP as of the market date. The decomposition is a matrix: columns are the book's three allocation slices (stable, non-stable, stress) and rows are the funding-credit legs (IR, TLP, HQLA buffer, all-in). The replicating weights come from the same frontier as the Replicating Portfolio tab, with the same engine toggle and corner selector.

A 30-day LCR stress runoff defines the stress slice, and the waterfall eats non-stable balances before stable. The stable slice earns its replicating-portfolio yield and TLP; the non-stable slice earns the three-month moving average of the three-month tenor and its 3M TLP; the stress slice earns the overnight rate (it is funded and held as overnight HQLA), carries no TLP, and bears the HQLA buffer charge. The buffer charge is the buffer target times the runoff times the cost of carry. The stock view uses realized trailing-MA carry; it is never interleaved with the marginal new-money view.

$$stressed = lcrRunoff, \quad nonStableSurviving = \max((1 - stableShare) - lcrRunoff, 0)$$

the runoff waterfall: stress depletes non-stable before stable

$$stableAfterStress = \max(stableShare - \max(lcrRunoff - (1 - stableShare), 0), 0)$$

only the spill beyond the non-stable slice reaches the stable slice

$$bufferCharge = bufferTarget \cdot lcrRunoff \cdot (carryBp / 1e4)$$

the HQLA cost-of-carry drag borne by the stress slice

$$allInCredit = irCredit + tlpCredit - bufferCharge, \quad ftpMargin = allInCredit - clientRate$$

client rate is 0 for Non-IB and the modeled D(0) for IB

- **Default assumptions.** Stable share 80%, LCR runoff 20% (Non-IB, treated as operational) or 40% (IB), buffer target 120%, cost of carry 15 bp. All are editable.
- **Stock vs marginal.** Stock legs use realized trailing-MA pillar carry; the marginal view prices the same slices at today's curve par. The two are kept on separate, labeled views.

Effective duration and economic value

Effective duration on the instrument tabs is a central finite difference of present value under a symmetric parallel shock, run as a real ALM EVE engine: only the market rate moves, while the behavioral cashflows and the discount factors are both re-derived from the shocked rates and every calibrated model parameter is held fixed. What makes it effective rather than modified is that the cashflows regenerate under each shifted path, so a genuine rate response propagates — mortgage prepayment re-fires, the deposit coupon reprices.

One design choice keeps the Non-IB regime amplifier freeze-free. The amplifier keys off the excursion $r - MA$: a permanent parallel shift moves r and its warmup-history MA by the same amount, leaving the excursion unchanged and the amplifier invariant. No freeze needed. The old cash-sorting salience (r/r_{ref} level multiplier) has been removed entirely in the current design, so there is nothing else that needs to be held at base. The EVE engine therefore re-levels the warmup history by the shock bp/100 on each shocked leg, which preserves the correct excursion without any special treatment.

Consequences differ by instrument. The Non-IB deposit pays a 0% coupon; its opportunity-cost incentive is a logistic in the spot rate LEVEL, so a +100bp shock slightly accelerates runoff and a –100bp shock slightly slows it — mild negative convexity. The decomposition hierarchy is WAL > static duration > effective duration: WAL is pure cashflow timing; static duration freezes the cashflows and shifts only the discount curve; effective duration lets both cashflows and discounting respond to the shock, compressing it further below static. The IB deposit reprices its coupon at $\beta(r) \cdot r$ and its opportunity-cost spread $r - D$ does move under the shift, so its runoff responds and its duration sits well below WAL (the repricing franchise), staying positive because the IB decay steepness is calibrated so the runoff response does not overwhelm the discounting. The curve shock shifts the zero curve directly ($DF \times \exp(-\Delta t)$); for a parallel move this matches shifting the par curve and re-bootstrapping to second order, and the par curve is what a bank actually transacts on, so the zero-shift is purely a computational simplification.

For premium or discount instruments an optional par anchor solves a single flat continuously-compounded spread on the base leg and holds it fixed across the shock legs; this is the option-adjusted-spread convention, and without it a deep-premium mortgage shows near-zero duration because the refinancing wave destroys the premium under the down-shock. The shock is applied from step 0 (the EVE convention) and discounting is on the all-in funding curve.

$$D_{eff} = (PV_- - PV_+) / (2 \cdot PV_0 \cdot \Delta r), \quad \Delta r = shockBp / 1e4$$

central difference; PV_0 , PV_+ , PV_- are path-mean present values at base and the two shocks

$$WAL = \Sigma_c (principalPaid_c / notional) \cdot (monthOffset_c / 12)$$

principal-runoff weighted-average life in years; interest excluded, no discounting

SEG / EBP

The sensitivity-equivalent gap measures the rate-locked outstanding balance of a static-balance-sheet portfolio through the sensitivity of net interest income to a small parallel shock. It converts a plus-or-minus ten basis point NII finite difference into an equivalent dollar notional that has not yet repriced, so the outstanding SEG profile is the option-adjusted, swap-equivalent repricing schedule of the book. For vanilla instruments it reproduces the repricing-gap line exactly; for convex instruments it bakes the linear swap-equivalent of the embedded option into the gap. The equivalent balance profile is the companion existing-book balance trajectory, with deterministic and Monte Carlo percentile fans.

Underneath sits a static-balance-sheet assumption: each month the existing book's runoff is replaced by a fresh vintage struck at the current shocked-rate environment, so total balance holds at notional. The carrying spread of each new vintage is referenced off the unshocked base curve on purpose, so the parallel shock propagates fully into the new coupon rather than being absorbed. The first month's coupon is locked before the shock (the NII convention), distinct from the EVE shock that moves step 0.

$$CumulativeSEG(t) = (NII_{up}(t) - NII_{down}(t)) \cdot 12 / \Delta r, \quad \Delta r = 2 \cdot SHOCK_BP / 1e4 = 0.0020$$

the central NII finite difference, annualized; $SHOCK_BP = 10$

$$OutstandingSEG(t) = side_sign \cdot (notional - CumulativeSEG(t))$$

the rate-locked outstanding balance; $side_sign$ is +1 for assets, -1 for liabilities

Glossary

All-in FTP — The transfer rate par-matched against the SOFR curve plus the term liquidity premium. The full funding cost the desk pays.

Antithetic pairing — A variance-reduction scheme that draws one shock per pair of paths and applies it with both signs, halving the variance of quantities roughly linear in the shock.

BGM / LMM — The LIBOR market model: a multi-factor forward-rate simulation calibrated to the swaption surface. One of the Lab's two rate engines.

Burnout — The decay of prepayment or deposit rate-sensitivity with cumulative in-the-moneyness, modeled as $\exp(-\text{rate} \cdot \text{cumulative incentive})$. Off by default for the deposit books.

CEV / displacement — The constant-elasticity-of-variance exponent and the additive shift applied to forwards in the BGM dynamics; structural controls on the upper tail, set exogenously.

Closure — The age-ramped baseline attrition of a deposit cohort, in percent per month, independent of rates; the floor under the rate-driven decay.

CPR / SMM — Conditional prepayment rate (annualized) and single-month mortality; $\text{SMM} = 1 - (1 - \text{CPR})^{1/12}$.

Deposit beta — The fraction of a market-rate move that passes through to the deposit rate, modeled as a logistic S-curve $\text{beta}(r)$; the IB deposit rate is $\text{beta}(r) \cdot r$.

Deposit tractor — The replicating portfolio built from moving-average pillars; a k-month pillar is k rolling bullets in the k-month tenor, WAL $(k+1)/2$ months.

EVE — Economic value of equity: the present value of assets minus liabilities, sensitive to the rate-shock applied. The Lab shocks from step 0 for EVE.

Effective duration — A central finite difference of PV under a permanent parallel shock, run as an EVE engine: cashflows and discount factors regenerate from the shocked rates while calibrated parameters are held fixed. Genuine rate response propagates — coupon repricing, prepayment, and the Non-IB opportunity-cost runoff (which shortens slightly as rates rise, a mild negative convexity with the operating level on the saturated plateau). The regime amplifier keys off the excursion $r - \text{MA}$, invariant to a parallel shift, so it needs no freeze.

FTP margin — The locked net interest margin: asset yield minus all-in FTP for assets, or all-in FTP credit minus deposit cost for liabilities.

g(b) balance sensitivity — The logistic-in-log-balance multiplier $g_{\min} + (g_{\max} - g_{\min}) / (1 + \exp(-\kappa_b (\ln b - \ln b^*)))$ that grades deposit rate-sensitivity by account size.

HQLA buffer charge — The cost-of-carry drag on the stress slice of the NMD FTP: buffer target times stressed runoff times the cost of carry.

Hull-White — A one-factor short-rate model with an exact OU transition and a martingale correction. One of the Lab's two rate engines.

IR FTP — The interest-rate component of FTP, par-matched against the SOFR curve alone, before any liquidity premium.

LCR runoff — The 30-day stressed outflow assumption from the liquidity coverage ratio, used to carve the contingent-liquidity stress slice out of the NMD book.

LP FTP — The liquidity-premium component of FTP: all-in FTP minus IR FTP. Additive by construction.

Martingale correction — A multiplicative per-step rescale of simulated discount factors so their path-mean equals the market discount factor at every horizon, removing discretization drift.

Nerlove partial adjustment — A geometric-lag smoothing of the deposit rate toward its target, $D(t) = D(t-1) +$

$\lambda \cdot (\text{target} - D(t-1))$; $\lambda = 1$ is full pass-through.

NMD — Non-maturity deposit: checking, savings, money-market balances with no contractual maturity. Modeled behaviorally.

OAS — Option-adjusted spread: the flat spread over the simulated discount paths that prices an instrument to par, capturing the embedded option.

Par-match — Solving the transfer rate that prices a cashflow stream to par on a given discount curve; linear in the rate, so a direct algebraic solve.

Pillar — A k-month moving-average building block of the replicating portfolio: k equal rolling bullets in the k-month tenor.

Rebonato volatility — The four-parameter (a, b, c, d) parametric instantaneous-volatility form of BGM, with a rotation angle beta setting the two-factor correlation.

Replicating portfolio — A blend of moving-average pillars whose cashflow and yield profile stands in for the behavioral deposit balance.

SABR — The stochastic-volatility smile model; here the Bachelier ($\beta = 0$) normal-vol expansion used to display the volatility surface.

Regime amplifier — The Non-IB surge multiplier on the rate-incentive limb, $1 + \alpha \cdot \max(0, r - \text{MA})$: when the spot rate exceeds its trailing moving-average regime, the low-yield deposits accumulated in the prior low-rate era reprice out faster. One-sided and dormant at or below the regime; the SSRN excursion form (Chen). It replaced the earlier rate-level cash-sorting multiplier and, keyed off the excursion, is invariant to a parallel EVE shock so it needs no freeze. The IB book uses no such multiplier — its beta coupon already prices in the rate level.

Seasoned stock vs cohort — The age-superposed deposit book versus a single fresh cohort; the stock is longer-lived because closure declines with age and survivors are stickier.

SEG / EBP — Sensitivity-equivalent gap and equivalent balance profile: the rate-locked outstanding balance from the NII sensitivity, and the existing-book balance trajectory.

TLP — Term liquidity premium: the spread of FHLB advance rates over matching-tenor SOFR. Zero at the overnight bucket by convention.

Vol value — The option premium a single deterministic rate path misses, measured as the stochastic minus deterministic FTP increment.

WAL — Weighted-average life: the principal-weighted average time to repayment of a cashflow stream.

Z-spread — The static spread over the deterministic discount curve that prices a cashflow stream to par.

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